

Mass ratio • Δv vs engine

WET-TO-DRY MASS RATIO m_0/m_f • ROUNDED TO NEAREST INTEGER

$$m_0 / m_f = e^{\Delta v / v_e} \cdot v_e = I_{sp} \cdot g_0$$

ratio > 50 marked – impossible
(unbuildable single stage)

Columns = Δv in km/s • merged cells share the same rounded ratio

engine ↓ Δv →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
● V-2 225 s • ethanol/LOX	2	4	6	10	15	24	38	–										
● Shuttle SRB 250 s • solid	2	3	5	8	12	17	26	39	–									
● F-1 (Saturn V) 263 s • kerolox	1	2	3	5	7	10	15	22	33	48	–							
● Merlin 1D 311 s • kerolox	1	2	3	4	5	7	10	14	19	27	37	–						
● Raptor 2 363 s • methalox	1	2	3	4	5	7	9	13	17	22	29	38	–					
● RS-25 (SSME) 452 s • hydrolox	1	2	3	4	5	6	8	10	12	15	19	24	29	37	46	56	67	79
● NERVA 841 s • nuclear thermal	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
● NEXT-C 4190 s • ion	1																	2
● VASIMR 5000 s • plasma	1																	

Cell color = engine (matches the lines on the Δv map). **Merged cells** share the same rounded ratio. The hatched region is where the mass ratio exceeds 50:1 – past that point a single stage is effectively unbuildable, since tanks, structure, and engine alone weigh more than 1/50 of liftoff mass. Real rockets beat the hatching only by staging: dropping dead weight resets the ratio mid-flight.

How to read: pick your mission Δv along the top, drop to your engine's row. A colored cell = feasible in one stage. Hatched = needs staging or a better engine. Note the cliff edge sweeping left as engines get worse: VASIMR never hits it, NERVA never hits it, but the V-2 goes impossible at just Δv 9 – it could never have reached orbit, no matter how it was built.